

SMART DISPLAY MODULE SPECIFICATION

4.3 Inch Smart Display with TOUCH	
Model:	PLCM-UES31S043H800V480C-U
MCU:	ESP32-S31-WROOM-3
PCB:	UES31S043H800V480C-U
Display:	4.3" IPS, 800 × 480, ST7262, RGB666/888, GT911 CTP
Version:	V1.0
Date:	2026-9-3

Customer Confirmation

Approved by	Notes

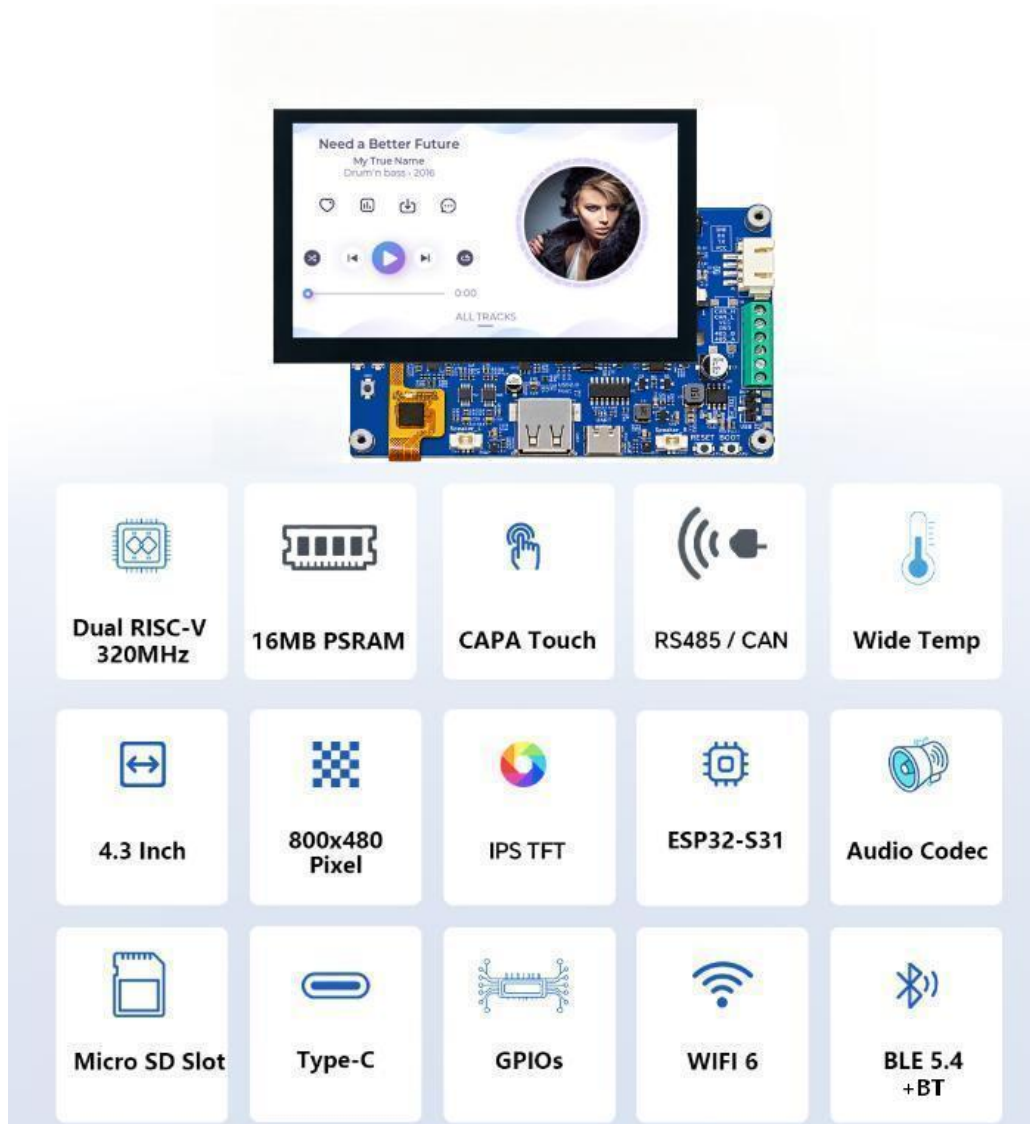
REVISION HISTORY

Revision	Date	Contents of Revision Change	Remark
V1.0	20260903		

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1. Introduction



PLCM-UES31S043H800V480C-U is a high-performance smart display module designed by VIEWE. It is based on the Espressif ESP32-S31-WROOM-3 module and a 4.3-inch RGB capacitive touch panel (800 × 480). The LCD and touch devices are the same as those used on UEDX80480043E-WB-B (ST7262 driver IC and GT911 CTP). The main board is a new design around ESP32-S31: a dual-core 32-bit RISC-V MCU running at up to 320 MHz, with Wi-Fi 6, Bluetooth 5.4 (LE) plus Bluetooth Classic, USB 2.0 High-Speed OTG, stereo audio, SDIO 3.0 storage, and RS485 / CAN industrial interfaces.

Notice: ESP32-S31 is a preview device. ESP-IDF builds must use the --preview option (for example: idf.py --preview set-target esp32s31). Module ratings follow the Espressif ESP32-S31-WROOM-3 datasheet (pre-release). New designs that need USB 2.0 High-Speed, Wi-Fi 6, or on-board codec audio should use this S31 board rather than the ESP32-S3.

1.1 Product Features

CPU:

1. Processor

- 1) ESP32-S31 RISC-V 32-bit dual-core processor, main frequency up to 320 MHz, plus a ULP-RISC-V coprocessor.
- 2) 2.4 GHz Wi-Fi 6 (IEEE 802.11b/g/n/ax, HT20/40, up to 150 Mbps), Bluetooth 5.4 LE (including LE Audio), Bluetooth Classic (BR/EDR), and IEEE 802.15.4 (Zigbee / Thread).
- 3) On-module PCB antenna.
- 4) Hardware security: Secure Boot, Flash/PSRAM encryption, cryptographic acceleration, and TEE. See the chip datasheet for details.

2. Memory

- 1) On-chip: 320 KB ROM, 512 KB SRAM, 32 KB low-power SRAM.
- 2) Typical module configuration on this product: 16 MB Quad SPI Flash + 16 MB Octal SPI PSRAM (ESP32-S31-WROOM-3-N16R16V).
- 3) Flash and PSRAM can be accessed in parallel.

3. Peripheral Interfaces

- 1) USB Type-C port for 5 V power, firmware download, and serial debug (CH340C bridged to UART0).
- 2) USB Type-A port connected to the dedicated USB 2.0 High-Speed OTG PHY (USB_DP / USB_DM, not GPIOs). Host 5 V is sourced through TPS2051C (~500 mA). VBUS-EN is hard-wired high and is not MCU-controlled.
- 3) On-board MicroSD slot, 4-bit SDMMC / SDIO 3.0.
- 4) ES8389 stereo codec, dual NS4150B speaker amplifiers, dual analog microphones, and L/R speaker connectors.
- 5) RS485 (SIT3088E, hardware automatic direction) and CAN (SIT1050T + on-chip TWAI) on a 6-pin terminal.
- 6) 2 × 10 pin 2.54 mm header (H1) bringing out UART0/UART1, I2C, RS485/CAN GPIOs, 3.3 V / 5 V / GND.
- 7) Four ADC keys, RESET / BOOT, WS2812B RGB LED, and a 3 kHz passive buzzer (Not attached by default) .

For more information on ESP32-S31-WROOM-3, please refer to the Espressif S31 datasheet .

Display

- 1) Size: 4.3 Inch
- 2) Resolution: 800 *480
- 3) Pixel Arrangement: RGB Vertical Stripe
- 4) Interface Mode: 40PIN RGB 24bits
- 5) Driver IC: ST7262E43-G4

- 6) Touch IC: GT911
- 7) Brightness: 400 cd/m²
- 8) Touch: CTP
- 9) Verified timing: PCLK 18 MHz, negative polarity; HSYNC 1 / 40 / 20; VSYNC 1 / 10 / 5.

More information about Display can be found here: [Display Specification.pdf](#)

Other

- 1) Operation Temperature: -20 ~ 70 ° C (limited by the LCD).
- 2) Storage Temperature: -30 ~ 80 ° C.
- 3) MCU module rating: -40 ~ 85 ° C.

1.2 Applications

With rich connectivity and powerful processing capabilities, the UEDX80480043E-WB-B is an ideal choice for IoT devices in the following areas:

- Smart Home Control Panels
- Industrial Automation HMI
- Smart Appliances
- Consumer Electronics
- Wireless Data Loggers
- Touch Screen Interfaces
- Educational Learning Platforms

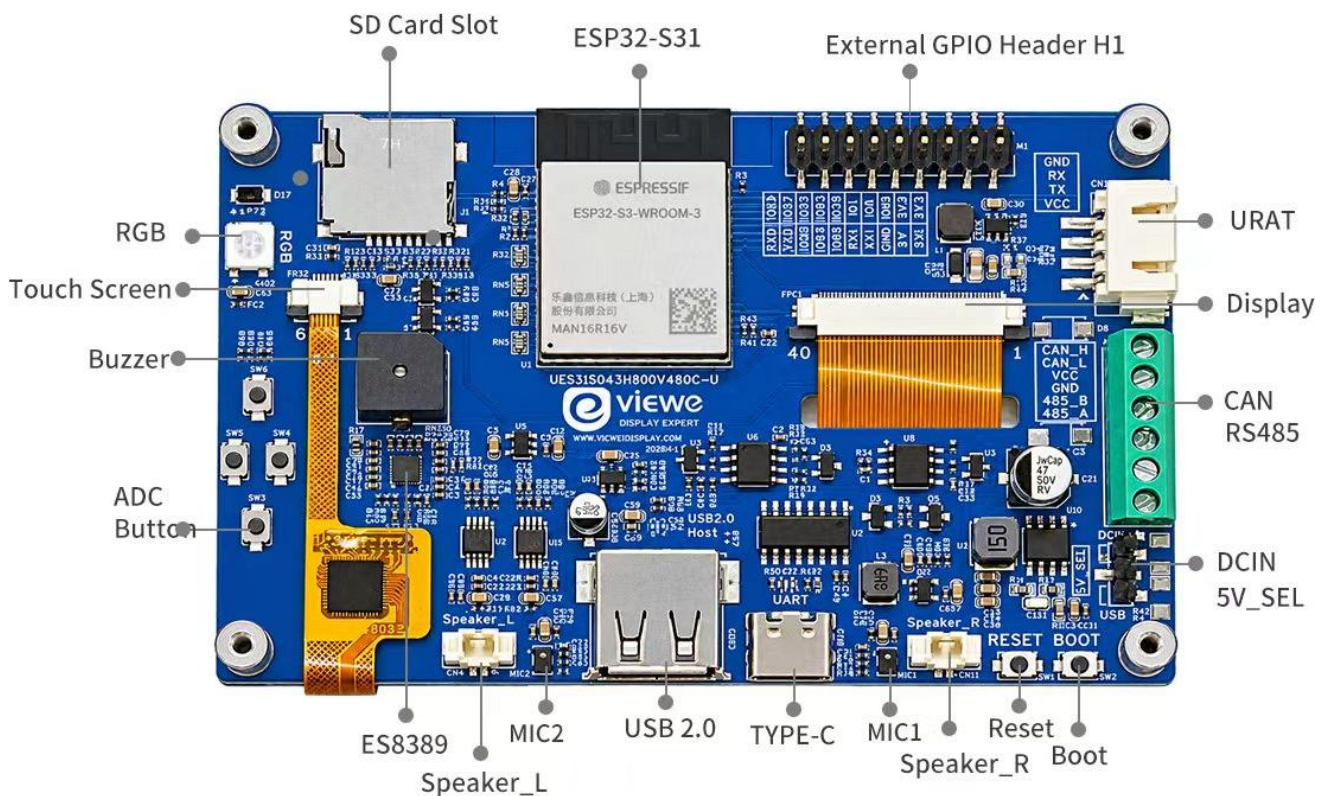
1.3 Product Naming

Field	Code	Meaning
Form	PLCM	PCB+LCM(Display)
Brand / series	UE	VIEWE smart display module
MCU	S31	ESP32-S31-WROOM-3
Size	S043	4.3 inch
Resolution	H800V480	800 × 480
Touch	C	Capacitive (GT911)
Bus	U	UART/RS485/I2C/CAN/USB 2.0 High-Speed OTG

2. Product information

2.1 Interface Description

This module achieves efficient communication with the main control unit via UART, RS485 and CAN buses, and supports USB 2.0 High-Speed OTG functions to meet diverse data interaction requirements. The capacitive touch screen adopts the GT911 solution to deliver smooth and responsive touch experience. The 4.3-inch screen features an 800×480 high-definition resolution, ensuring clear and delicate display content. Integrated with the ESP32-S31-WROOM-3, it balances high-performance processing and low-power operation. Suitable for various industrial and commercial application scenarios, it provides stable and reliable visual interaction support for devices.



1、 Main Control Module:

ESP32-S31-WROOM-3 Dual-core RISC-V, up to 320 MHz, 16 MB Flash + 16 MB PSRAM, PCB antenna.

2、 Display Interface:

40-pin RGB output. The panel is 24-bit capable; this board wires RGB666 (R2–R7 / G2–G7 / B2–B7). See the Display Interface table.

3、 SD Card Slot:

SPI interface connected for external storage expansion, Please refer to Section [2.9](#) below for detailed information.

4、 Touch interface:

I2C (SDA = GPIO0, SCL = GPIO1) to GT911. INT and RST are present on the 6-pin FPC; verified examples leave them unused (NC).

5、 USB Type-C interface:

5 V DC input, programming, and serial debug through CH340C

6、 USB Type-A:

USB 2.0 High-Speed OTG PHY (USB_DP / USB_DM). Host 5 V via TPS2051C. VBUS-EN is pulled high by hardware. Please refer to Section [2.7](#) below for detailed information.

7、 UART helper (4-pin):

GND / RX / TX / VCC for an external serial adapter. The 4-pin pitch of 2.5 mm

8、 RGB-LED (WS2812B):

one WS2812B, DIN = GPIO37, powered from 5 V with a level-shift network. Please refer to Section [2.9](#) below for detailed information.

9、 Boot Buttons:

BOOT button (GPIO61) for firmware download mode.

10、 Reset Buttons:

RESET button (CHIP-EN) for resetting the board.

11、 External GPIO header H1:

2 × 10 pins, 2.54 mm. Please refer to Section [2.6](#) below for detailed information.

12、 Industrial terminal CN3 (CAN/RS485):

CAN_H, CAN_L, VCC, GND, 485_B, 485_A, Please refer to Section [2.8](#) below for detailed information.

13、 Audio:

Dual-channel speaker interface Speaker L / R (CN2 / CN1) with a 2-pin pitch of 1.25 mm, and on-board dual microphones MIC1 / MIC2 deliver high-quality audio via ES8389 and NS4150B, Please refer to Section [2.5](#) below for detailed information.

14、 ADC Button:

ADC key array SW3–SW6 on GPIO42. Please refer to Section [2.9](#) below for detailed information.

15、 Buzzer:

BEEP_EN = GPIO46, high = on, 3 kHz passive type. Not attached by default

16、 DCIN / 5V_SEL:

external 5 V input and 5 V source-selection jumper.

Display Interface:

Pin No.	Symbol	I/O	Description
1	LEDK	P	Power supply for backlight cathode
2	LEDA	P	Power supply for backlight anode
3	GND	P	Power ground
4	VDD	P	Logic supply, 3.3 V
5-12	R0-R7	I	Red data. This board uses R2–R7 (GPIO2–7). R0/R1 = NC
13-20	G0-G7	I	Green data. This board uses G2–G7 (GPIO8–13). G0/G1 = NC
21-28	B0-B7	I	Blue data. This board uses B2–B7 (GPIO14–19). B0/B1 = NC
29	GND	P	Power ground
30	CLK	I	Pixel clock, negative polarity (GPIO40)
31	DISP	I	Standby mode. Normally pulled high
32	HSYNC	I	Horizontal sync, negative polarity (GPIO44)
33	VSYNC	I	Vertical sync, negative polarity (GPIO45)
34	DEN	I	Data enable. Display access is enabled when DE is “H” (GPIO43)
35	NC	I	Dummy
36	GND	P	Power ground
37	XR	-	Dummy
38	YD	-	Dummy
39	XL	-	Dummy
40	YU	-	Dummy

I: Input; O: Output; P: Power

TP Interface:

Pin No.	Symbol	I/O	Description
1	RST	P	Touch reset on FPC; not used
2	3.3V	P	3.3 V logic supply
3	GND	P	Power ground
4	INT	I	Touch interrupt on FPC; TP INT = GPIO38
5	SDA	I	SDA = GPIO0, shared with ES8389
6	SCL	P	SCL = GPIO1, 400 kHz, shared with ES8389

2.2 GPIO Definition

GPIO Number	Current Usage	Function Description
CHIP-EN	Reset	RESET button (SW1)
GPIO0	I2C SDA	GT911 + ES8389. Pulled high at reset (normal boot). Not the BOOT key; also on H1
GPIO1	I2C SCL	GT911 + ES8389; also on H1
GPIO2–7	LCD-R2–R7	LCD red data, high 6 bits
GPIO8–13	LCD-G2–G7	LCD green data, high 6 bits
GPIO14–19	LCD-B2–B7	LCD blue data, high 6 bits
GPIO20–23	SD-D0–D3	SDMMC 4-bit data
GPIO24	SD-CLK	SDMMC clock
GPIO25	SD-CMD	SDMMC command
USB_DP / USB_DM	USB 2.0 HS PHY	Dedicated module pins. Do not configure as GPIO
GPIO33	UART1 TX	Verified as UART1 TX. Also a USB Serial/JTAG pad
GPIO34	UART1 RX	Verified as UART1 RX
GPIO35	RS485 TX	SIT3088E DI; S8050 auto-drives DE/~RE; also on H1
GPIO36	RS485 RX	SIT3088E RO; also on H1
GPIO37	WS2812 DIN	On-board XL-5050RGB-WS2812B
GPIO39	LCD-BL-EN	Backlight enable, active high
GPIO40	LCD-PCLK	Pixel clock, 18 MHz, negative polarity
GPIO42	ADC KEY	SW3–SW6 resistor ladder, ADC1, ~0–2 V usable range
GPIO43	LCD-DE	Data enable
GPIO44	LCD-HS	Horizontal sync (GPIO number, not USB_DM)
GPIO45	LCD-VS	Vertical sync (GPIO number, not USB_DP)
GPIO46	BEEP-EN	Buzzer, active high; 3 kHz PWM recommended
GPIO47	PA-EN	NS4150B amplifier enable, active high
GPIO48	I2S MCLK	Present on schematic; firmware examples leave NC (codec no_mclk)
GPIO49	I2S BCK	ES8389 bit clock
GPIO50	I2S WS	ES8389 word select / LRCK
GPIO51	I2S DOUT	MCU → ES8389 DAC
GPIO52	I2S DIN	ES8389 ADC → MCU
GPIO53	CAN TX	SIT1050T TXD
GPIO54	CAN RX	SIT1050T RXD
GPIO55–57	H1 GPIO	Broken out on H1
TX0 (IO58) / RX0 (IO59)	UART0	CH340C download / log; also on H1
GPIO60	SD-CTRL	SD 3.3 V power switch, active low
GPIO61	BOOT	BOOT button (SW2); also on H1

SD	I2C	Reset
RS485	UART1	USB2.0
Buzzer	LCD	RGB
CAN	I2S	PA-EN
GPIO	UART0	Boot

2.3 Display Information

Item	Specification	Unit	Remark
Pixel Driving element	IPS TFT	-	-
Screen Size	4.3	Inch	Diagonal
Resolution	800(W)*3(RGB)*480(H)	Dots	-
Interface	RGB 24bits	-	40PIN
Module Power Consumption	1.04	Watt	Typ.
Active Area	95.04(W)*53.86(H)	mm	-
Pixel pitch (W*H)	0.1188(W)*0.1122(H)	mm	-
Module Size (W*H*D)	105.52(W)*67.17(H)*2.8(D)	mm	Tolerance: ± 0.2
Luminance	400	cd/m ²	Typ.
Viewing Direction	All	O'clock	-
Display Color	16.2M	Colors	24bits

2.4 LCD RGB timing (verified)

Parameter	Value
PCLK	18 MHz, pclk_active_neg = true
Horizontal resolution	800
Vertical resolution	480
HSYNC pulse / back porch / front porch	1 / 40 / 20
VSNC pulse / back porch / front porch	1 / 10 / 5
Framebuffer	RGB888 in PSRAM, one frame buffer

2.5 Audio Interface

The audio codec is Everest ES8389 (I2C 7-bit address 0x20). I2S0 is used in full duplex. 48 kHz / 16-bit / stereo playback and record have been verified. MCLK is not required (codec configured with no_mclk). Dual NS4150B Class-D amplifiers drive the left and right speakers. PA_EN = GPIO47. MIC1 and MIC2 are on-board analog microphones.

Signal	GPIO / Device	Description
I2C SDA / SCL	GPIO0 / GPIO1	Shared with GT911, 400 kHz
I2S BCK	GPIO49	Bit clock
I2S WS	GPIO50	Word select / LRCK
I2S DOUT	GPIO51	DAC playback
I2S DIN	GPIO52	ADC record
I2S MCLK	NC in firmware	Optional on schematic
PA_EN	GPIO47	High = amplifier on
Speaker	CN1 / CN2	NS4150B differential output, typ. 4 Ω / 3 W class

2.6 H1 header (2 × 10, 2.54 mm), top to bottom

Number	Left	Number	Right
1	3V3	2	5V
3	3V3	4	5V
5	GND	6	GND
7	GPIO0 (SDA)	8	TX1 (GPIO33)
9	GPIO1 (SCL)	10	RX1 (GPIO34)
11	GPIO35 (485_TX)	12	GPIO36 (485_RX)
13	GPIO53 (CAN_TX)	14	GPIO54 (CAN_RX)
15	GPIO55	16	GPIO56
17	GPIO57	18	TX0
19	GPIO61 (BOOT)	20	RX0

GPIO35/36 and GPIO53/54 on H1 are in parallel with the on-board RS485 and CAN transceivers. Do not drive both the header and the terminal at the same time. GPIO0/1 already have I2C pull-ups.

2.7 USB 2.0 Interface

ESP32-S31 integrates a USB 2.0 High-Speed OTG PHY on dedicated USB_DP / USB_DM pins (package pins 44 / 45 of the chip). These are not GPIOs and need no GPIO-matrix setup. The board provides two physical connectors:

Port	Bridge / PHY	Role	Notes
Type-C	CH340C → UART0	Download / log / 5 V in	Hold BOOT (GPIO61), then tap RESET
Type-A	USB 2.0 HS PHY	OTG Host or Device	TPS2051C sources 5 V; VBUS-EN hard-wired high

Verified applications: (1) TinyUSB HID mouse mapped from the touch panel — class-compliant on Windows and Linux, no custom driver; (2) USB extended display — host-side IDD / vendor driver required, JPEG decode to the RGB panel. Type-A defaults to supplying 5 V as a USB Host. When the same port is used as a Device attached to a PC, avoid back-feeding VBUS from both ends.

2.8 UART / RS485 / CAN

Interface	Controller	MCU pins	External	Verified setting
UART0	CH340C	TX0 / RX0	Type-C / H1	Download and monitor
UART1	GPIO matrix	TX=33, RX=34	H1 TX1 / RX1	115200 8N1
RS485	SIT3088E + UART2	TX=35, RX=36	CN3 485_A / 485_B / GND	115200 8N1, auto DE
CAN	SIT1050T + TWAI	TX=53, RX=54	CN3 CAN_H / CAN_L / GND	500 kbit/s, Classic CAN

CN3 terminal (follow the PCB silkscreen)

Silk	Function
CAN_H	CAN high
CAN_L	CAN low
VCC	5 V — check jumper and load before using as a supply
GND	Common ground
485_B	RS485 B
485_A	RS485 A

RS485: DE/~RE is switched automatically from 485_TX through an S8050. Treat the port as a normal UART in software; do not configure RTS. 120 Ω termination and A-pull-up / B-pull-down are on board. CAN: 120 Ω termination on board; CAN_GND is tied to system ground through 0 Ω. Do not connect a USB-TTL adapter to A/B or CAN_H/L.

2.9 Storage, Keys, LED

Micro SD

SDMMC 4-bit: CLK = GPIO24, CMD = GPIO25, D0–D3 = GPIO20–23, SD_CTRL = GPIO60 (active low).

Examples mount FAT at SDMMC_FREQ_HIGHSPEED. Assert GPIO60 low before initializing the host.

ADC keys (measured on the real board)

Key	Typical voltage	Software window	Remark
Idle	≈ 3.3 V (pull-up)	Saturated / idle	S31 ADC 0 dB ≈ 0–2 V; idle saturates
SW3	≈ 0.38 V	100–600 mV	GPIO42
SW4	≈ 0.82 V	600–1080 mV	GPIO42
SW5	≈ 1.34 V	1080–1605 mV	GPIO42
SW6	≈ 1.87 V	1605–2200 mV	GPIO42

Two keys cannot be detected at once (inherent to the divider). Silkscreen order on some drawings does not match the measured voltages; use the table above.

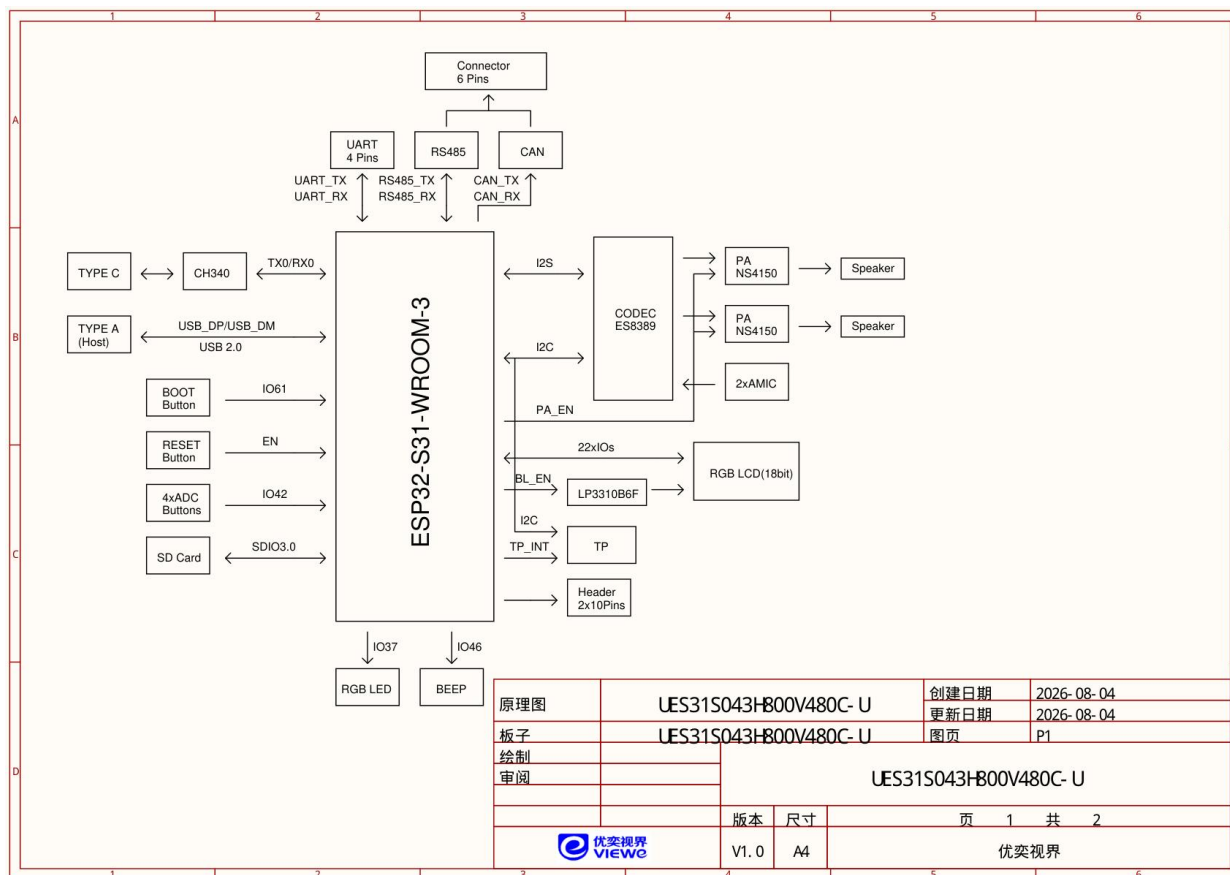
2.10 Voltage & Current

Item	Conditions	Min	Typ	Max	Unit
Power Voltage	DC	-	5.0	-	V
Operation Current	VCC= +5V, Maximum backlight current	80	320	500	mA
	VCC= +5V,backlight off	-	100	-	mA
Recommended power supply:5V 1A DC					

2.11 Reliability Test

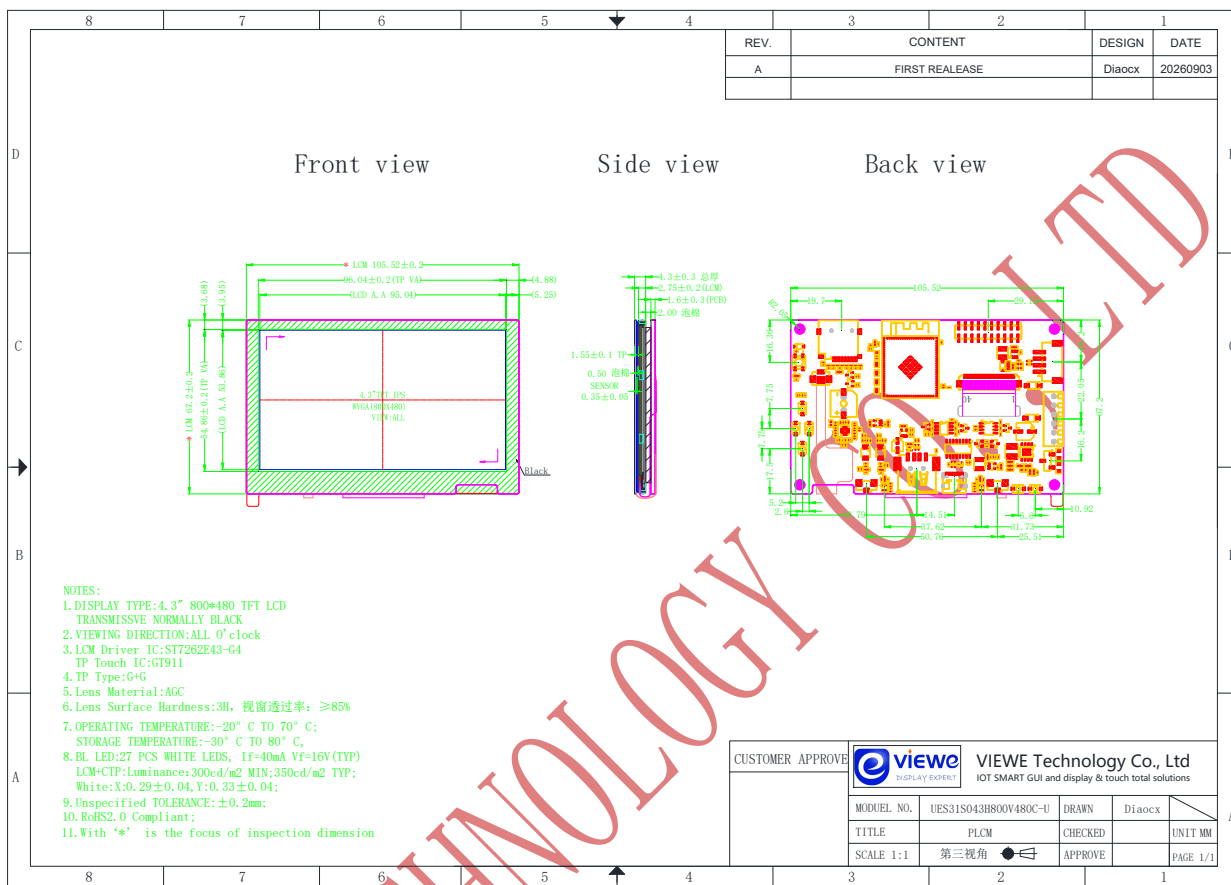
Item	Conditions	Min	Typ	Max	Unit
Working Temperature	60%RH at 5V voltage	-20	25	70	C
Storage Temperature	---	-30	25	85	C
Working Humidity	25°C	10%	60%	90%	RH
ESD	---	Contact: ±4KV Air: ±8KV			KV

3. Functional Block Diagram



Note: ESP32-S31 includes a 2.4 GHz radio that supports Wi-Fi 6, Bluetooth 5.4 (LE), Bluetooth Classic, and 802.15.4. Because they share the same RF front-end, Wi-Fi and Bluetooth cannot transmit or receive simultaneously; the radio switches between protocols as needed. USB 2.0 High-Speed is an independent PHY and does not share that front-end.

4. MECHANICAL DRAWING



5. Detailed information (codes, drawings...)

Github: <https://github.com/VIEWESMART/PLCM-UES31S043H800V480C-U>

Gitee: <https://gitee.com/VIEWESMART/PLCM-UES31S043H800V480C-U>

Wiki: <https://displaywiki.com/products/esp32-s31-4.3inch-800x480-touch-uart-display-hmi>

Official Store:

<https://viewedisplay.com/product/4-3-inch-800x480-esp32-s31-wifi6-touch-smart-hmi-uart-display>

Prompt: The materials are being developed and enriched. Please stay tuned.

VIEWE TECHNOLOGY CO., LTD